

SECTION 1: QUANTUM-RESILIENT ACCESS PROTOCOL

1.1 Purpose: This operational document establishes binding cryptographic controls for classified information provenance using National Institute of Standards and Technology (NIST) FIPS 203 (ML-KEM-768) and FIPS 204 (ML-DSA-65).

1.2 Trust Architecture: Secret cryptographic material remains strictly confined within authorized recipient browser enclaves. Zero key exposure or plaintext transmission occurs across server boundary components.

1.3 Dual-Key Secret Sharing: Plaintext PDF assets are symmetrically locked under AES-256-GCM file keys (K) generated dynamically. The file key is divided into K1 (held in the encrypted distribution bundle) and K2 (governed by the immutable consensus key release arbiter).

SECTION 2: COMMIT-BEFORE-RELEASE LEDGER PROVENANCE

2.1 Provenance Gate: A recipient cannot obtain key share K2 without executing an irrevocable, ML-DSA-65 signed decryption provenance record containing unique session tokens and client nonces.

2.2 Consensus Validation: A three-node permissioned ledger independently verifies all recipient signatures and cryptographic parameters, requiring a 2-of-3 Byzantine quorum before stamping block inclusion.

2.3 Forensic Watermark ID: The recipient browser computes a 96-bit cryptographic watermark identifier bound to doc_id, recipient_id, session_id, and mutual nonces, permanently anchored into the distributed ledger.

SECTION 3: FORENSIC ATTRIBUTION & INVESTIGATION DISCLOSURE

3.1 Dual-Channel Watermarking: Plaintext pages are transformed using 8x8 block discrete cosine transform (DCT) differential luminance modulation alongside structural XMP/invisible-text tokens.

3.2 Multi-Investigator Quorum: In accordance with chain-of-custody statutory requirements, forensic case extraction requires a minimum of two authorized investigator signatures before attribution data can be unlocked.

3.3 Tamper Evident Guarantee: Any retroactive alteration of provenance blocks triggers instant consensus divergence, maintaining incontrovertible evidence of legitimate recipient access.